
Title of the project: HYBRID CHAOTIC LIGHT IMAGE ENCRYPTION FOR SECURE UAV TRANSMISSION : WAVELET AND XOR-DIFFUSION APPROACH

Abstract:

The secure transmission of digital images has become a critical requirement in modern UAV communication systems due to the increasing risk of cyber threats targeting sensitive visual information. Images transmitted over wireless communication channels are vulnerable to interception, modification, statistical attacks, differential attacks, and unauthorized access. Existing encryption techniques often provide strong security but may involve high computational complexity, making them less suitable for real-time image transmission.

This project presents a Hybrid Chaotic Lightweight Image Encryption for Secure UAV Transmission using a Wavelet and XOR-Diffusion Approach, designed as a purely software-based encryption framework that combines computational efficiency with robust security. The proposed method employs the Discrete Wavelet Transform (DWT) to decompose the input image into multiple frequency components, enabling efficient image representation while preserving essential visual features. A chaotic map is utilized to generate highly unpredictable encryption keys, taking advantage of its randomness, nonlinearity, and extreme sensitivity to initial conditions. The generated chaotic keys are integrated with an XOR-based diffusion algorithm to perform pixel-level encryption, effectively reducing pixel correlation and increasing resistance against cryptographic attacks.

The proposed encryption framework provides multiple security advantages, including a large key space, high information entropy, strong key sensitivity, low pixel correlation, and resistance to brute-force, statistical, differential, replay, and interception attacks. The performance of the proposed approach is evaluated using standard image encryption metrics such as Information Entropy, Histogram Analysis, Correlation Coefficient, NPCR (Number of Pixel Change Rate), UACI (Unified Average Changing Intensity), PSNR (Peak Signal-to-Noise Ratio), MSE (Mean Square Error), Encryption Time, Decryption Time, and Key Sensitivity Analysis.